Freight Rate Machine Learning Report

FUTURE-AWARE FREIGHT RATE REGRESSION

Machine Learning Engineering Assessment

Prepared October 2026

Development	Prediction set	Jul-Oct backtest	Final output
48,000 labeled	12,000 future	$114.20 MAE	12,000 valid rates

Executive Summary

This assessment develops a reproducible regression pipeline for posted freight rates. Development covers January through October 2025 and final inference covers November and December, so model selection uses expanding-window monthly holdouts rather than random cross-validation.

The selected CatBoost RPM model achieved $114.20 mean temporal MAE, 4.78% WAPE, and $42.41 mean median absolute error at 400 iterations. It improved mean MAE by approximately 17.7% over the business baseline.

- Negative and missing weights are repaired with explicit anomaly indicators.
- All learned preprocessing values are fit inside each temporal training fold.
- quote_signal is excluded after temporal stability analysis and ablation testing.
- Eight-city cold-start simulations test geographic generalization.
- All 12,000 final predictions and 31 December rows pass the supplied scorer.

1. Objective and Data

The objective is to estimate posted_rate for every final validation load and generate a daily December curve for a fixed Lexington-to-Fort Wayne dry-van scenario. The solution must generalize across future dates, market conditions, new cities, and unseen route combinations.

Dataset	Rows	Period	Role
Development	48,000	2025-01-01 to 2025-10-31	Training and temporal validation
Validation	12,000	2025-11-01 to 2025-12-31	Final inference
December	31	2025-12-01 to 2025-12-31	Fixed-load scenario

2. Exploratory Analysis and Data Quality

The target mean is $2,373.98 and median is $2,030.76. A conservative RPM IQR rule identifies 663 extreme rows (1.38%). Tested configurations predict this segment poorly. Possible contamination is one hypothesis alongside legitimate rare events and omitted predictors.

Issue	Development	Validation	Treatment
Missing weight	300	165	Equipment median + indicator
Negative weight	292	145	Absolute value + indicator
Missing market	374	249	Fold median + indicator
Extreme RPM	663	Unknown	Robust MAE objective

Validation adds 8 new cities (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego) and 87.83% of its rows use a route observed in development. The market mean falls from 1.083 to 0.927. quote_signal has only -0.014 correlation with RPM in August and alternates sign in other months.

3. Preprocessing and Feature Engineering

- Drop load_id and separate posted_rate before feature transformation.
- Repair weight signs; retain missing and negative indicators.
- Apply fold-local weight and market imputations.
- Create route, log-distance, coordinate-delta, haversine, and bearing features.
- Create linear and cyclical calendar features.
- Preserve pickup, delivery, equipment, and route for native CatBoost handling.

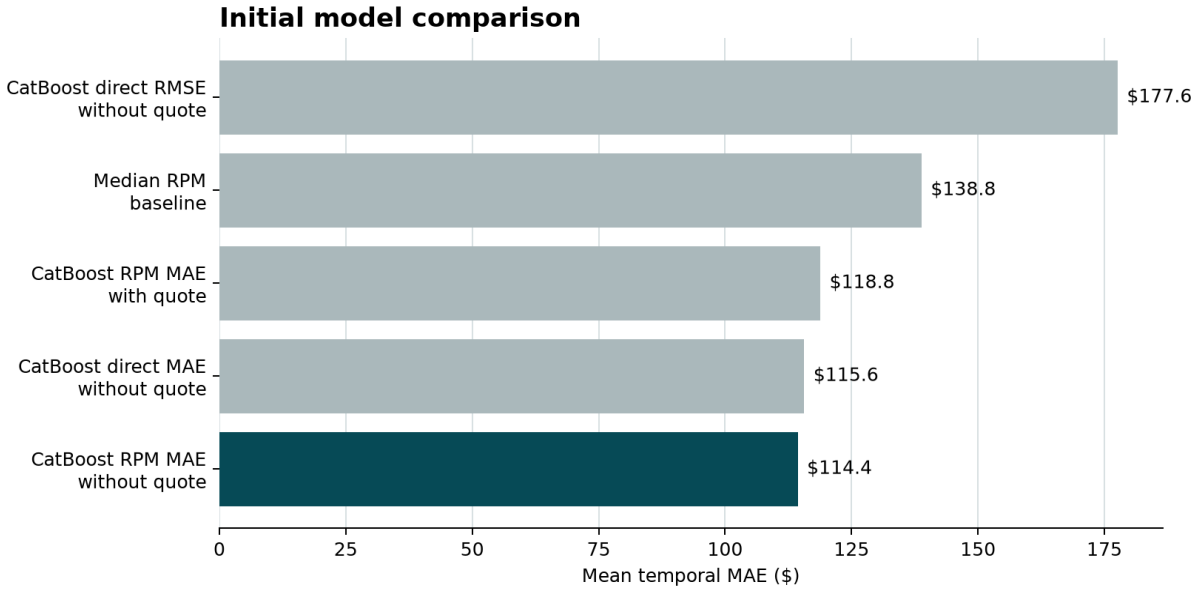
4. Validation Strategy

Four expanding-window folds mirror the production horizon. MAE is primary; RMSE, median absolute error, WAPE, RMSLE, bias, outlier performance, and route coverage are diagnostics.

Fold	Training window	Validation
1	January-June	July
2	January-July	August
3	January-August	September
4	January-September	October

5. Model Selection

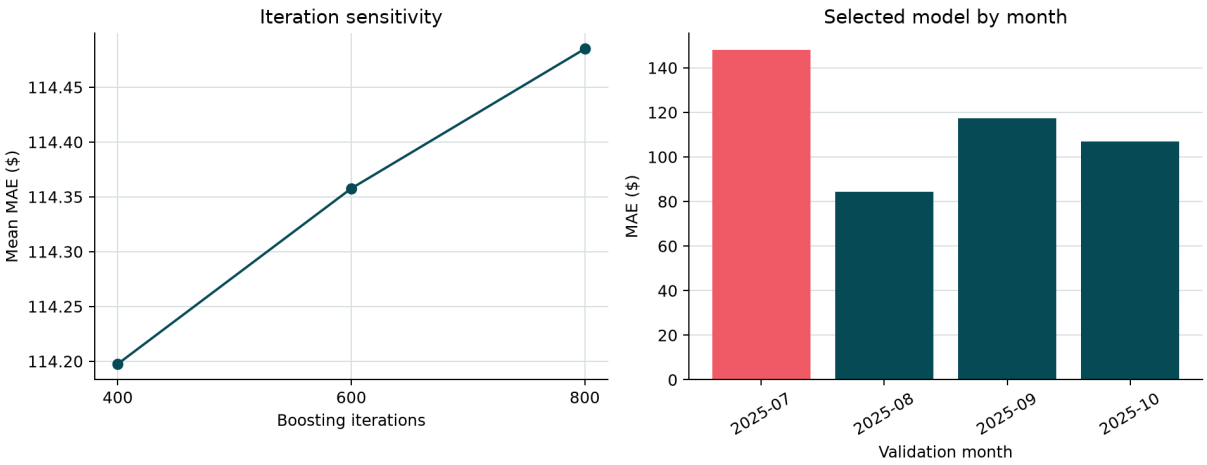
Candidate	Mean MAE	WAPE	RMSE
CatBoost RPM MAE, no quote	$114.36	4.78%	$627.48
CatBoost direct MAE, no quote	$115.63	4.84%	$627.65
CatBoost RPM MAE, with quote	$118.82	4.98%	$629.99
Median RPM baseline	$138.84	5.82%	$634.73
CatBoost direct RMSE, no quote	$177.60	7.42%	$657.94



Predicting RPM captures freight economics. Distance weighting aligns RPM MAE with dollar MAE after reconstruction. The quote ablation supports exclusion for future inference.

6. Robustness Checks

Iterations	Mean MAE	Worst MAE	Runtime
400	$114.20	$148.13	126 s
600	$114.36	$150.70	189 s
800	$114.48	$151.57	254 s



Four hundred iterations had the best mean and worst-month MAE. In three eight-city cold-start simulations, retaining route achieved $146.70 mean MAE versus $149.35 without route.

7. Final Model and Performance

Component	Choice	Component	Choice
Estimator	CatBoost	Target	posted_rate / distance
Objective	MAE	Weight	distance
Iterations	400	Depth	7
quote_signal	Excluded	Route	Included

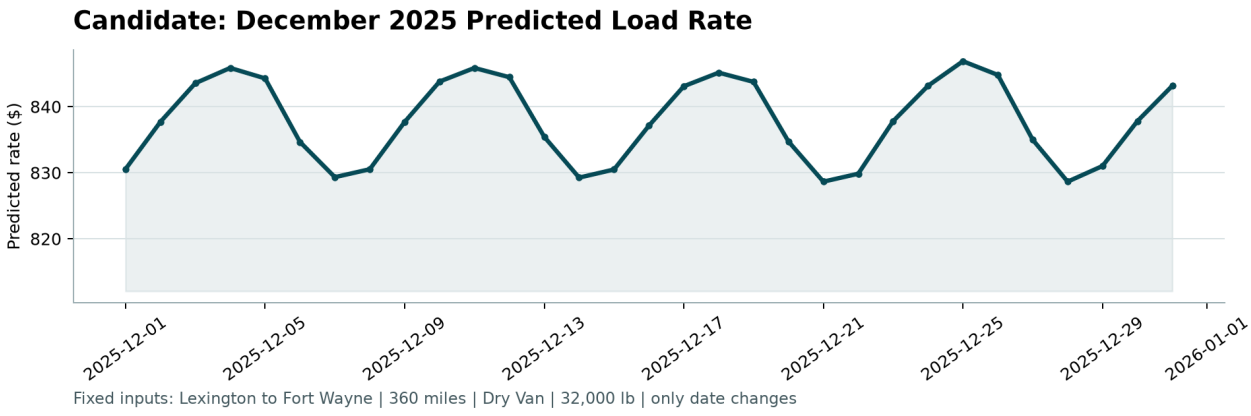
Mean MAE	Worst month	Median AE	WAPE	Bias
$114.20	$148.13	$42.41	4.78%	-$21.33

Regular-target MAE is $61.02; MAE for rows extreme under the conservative RPM rule is $3,880.05. This segment drives the $627.62 RMSE and is predicted poorly by tested configurations; possible contamination is one hypothesis alongside legitimate rare events and omitted predictors.

8. Final Inference and December Scenario

The model was refit on all 48,000 labeled rows and produced 12,000 scorer-valid predictions from $185.03 to $6,833.10. All IDs are unique and all rates are finite and positive.

For the fixed December scenario, coordinates come from the labeled city mapping and each date uses the daily mean market index from supplied unlabeled validation covariates. quote_signal is excluded from the final model.



Scorer-generated December curve: $828.64-$846.87, mean $837.86.

9. Limitations and Production Considerations

- The hidden scoring metric and acceptance threshold are not specified.
- Approximately 1.4% of development labels are extreme under the conservative RPM rule; tested configurations predict this segment poorly. Possible contamination is one hypothesis alongside legitimate rare events and omitted predictors.
- Historical backtest error is concentrated in RPM-extreme rows; final-period novelty risk is unmeasured because its labels are hidden.
- December market enrichment should become an explicit market-data feed in production.
- Production monitoring should track drift, route coverage, prediction ranges, and monthly segment error.

10. Reproducibility

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
.\.venv\Scripts\python.exe -m src.train --iterations 600
.\.venv\Scripts\python.exe -m src.robustness
.\.venv\Scripts\python.exe -m src.predict --iterations 400
.\.venv\Scripts\python.exe score.py --predictions validation_predictions.csv --december-predictions
december_predictions.csv
```

Environment: Python 3.12.3, pandas 2.3.3, NumPy 2.5.3, scikit-learn 1.9.1, CatBoost 1.2.10. Ten automated tests and the supplied scorer passed.

Conclusion

The final pipeline combines future-aware validation, robust loss, domain feature engineering, explicit instability tests, and scorer-validated outputs. Performance is strong for ordinary loads. Historical backtest error is concentrated in RPM-extreme rows, while final-period novelty risk remains unmeasured.